SkyGuard Fault Detection

Neil Mittal Roumya Lenka Korla Bharat
Mahin Mandava Tanmay Khandelwal Aadya Sisodya
Team Suiiii

This report summarizes the procedure followed in building a faulty flight detection algorithm for SkyGuard. The problem is a classification task built on three different modalities, each with its own use case and contribution to the final prediction. For each modality we performed exploratory data analysis to look for underlying patterns, followed by feature extraction and model building using classical machine learning algorithms.

1. Tabular Dataset

PCA and t-SNE showed no clear pattern in lower dimensions, so no strong assumption could be made about the data from these alone. For feature engineering, target encoding was used for the model name (applied during fusion), while one-hot encoding was used for the remaining categorical features. The data was then scaled, and group cross-validation was conducted with $n = 6$ splits grouped by `drone_id`.

We trained several classical approaches: random forest, logistic regression, and XGBoost. Surprisingly, the highest Average Precision was achieved by logistic regression (32%), followed by random forest (29%), with decision trees performing worst (15%).

2. Notes Dataset

The maintenance notes provide valuable unstructured information recorded after each flight. During exploratory analysis, we observed that notes associated with faulty flights frequently contained terms related to battery issues, vibrations, overheating, and component replacements, while healthy flights generally contained routine inspection remarks or no notes at all.

As a baseline, maintenance notes were converted into numerical feature vectors using TF-IDF with unigrams and bigrams, replacing missing notes with empty strings to retain all training samples. Using these features with a Logistic Regression classifier achieved an Average Precision (AUPRC) of 42%, showing that maintenance logs contain meaningful predictive information and are a valuable modality for failure prediction.

3. Signals Dataset

Comparing time series of a healthy and a faulty flight showed that healthy flights followed a more predictable pattern with lower magnitude in motion variables, while faulty flights showed larger variance and mean.

Initially, each 128×8 signal subset was compressed into a single row per flight ID using statistical features: minimum, maximum, mean, variance, RMS value, median, skew, and kurtosis. This gave useful insights. Some features showed similar patterns across healthy and faulty flights, while others showed a clear distinction. Analyzing more such instances could help identify the root cause of failure in the drones.

We later added features such as mean amplitude, energy, max amplitude, and dominant frequency obtained from a Fast Fourier Transform of each flight's time series, with the baseline amplitude at frequency zero set to zero. The maximum accuracy reached with these signal features alone was around 32% with logistic regression, followed by 28% with random forest.

4. Final Fusion

After developing individual baselines, the tabular, signal, and text features were fused into a single feature vector for each flight. The tabular features captured operational characteristics, the signal features summarized temporal sensor behavior through statistical descriptors, and the TF-IDF vectors represented information from the maintenance notes.

For categorical variables, One-Hot Encoding was used for low-cardinality features, while Target Encoding was applied to the model feature to handle the unseen Model E in the test set and avoid the sparsity introduced by One-Hot Encoding. Hyperparameter tuning was performed using `RandomizedSearchCV` with `GroupKFold` cross-validation grouped by `drone_id` to prevent data leakage.

The tuned fusion model achieved a final Average Precision (AUPRC) of **51.4%** on Logistic Regression, demonstrating the effectiveness of combining complementary information from all three modalities.